

变限积分函数求导-练习题解答

1. 设函数 $y = y(x)$ 由方程 $\int_1^y e^{t^2} dt - \int_0^x \cos t^2 dt = \sin x$ 所确定, 求 $\frac{dy}{dx}\bigg|_{x=0}, \frac{d^2y}{dx^2}\bigg|_{x=0}$.

解 方程两边关于 x 求导得

$$e^{y^2} \frac{dy}{dx} - \cos x^2 = \cos x$$

再求导得

$$e^{y^2} \cdot 2y \cdot \left(\frac{dy}{dx}\right)^2 + e^{y^2} \frac{d^2y}{dx^2} + \sin x^2 \cdot 2x = -\sin x$$

在原方程中令 $x = 0$ 得 $\int_1^y e^{t^2} dt = 0 \Rightarrow y = 1$, 将 $x = 0, y = 1$ 代入前面第一式得

$$e \frac{dy}{dx}\bigg|_{x=0} - 1 = 1 \Rightarrow \frac{dy}{dx}\bigg|_{x=0} = \frac{2}{e}$$

将 $x = 0, y = 1, \frac{dy}{dx} = \frac{2}{e}$ 代入前面第二式得

$$e \cdot 2 \cdot \left(\frac{2}{e}\right)^2 + e \frac{d^2y}{dx^2}\bigg|_{x=0} = 0 \Rightarrow \frac{d^2y}{dx^2}\bigg|_{x=0} = -\frac{8}{e^2}$$

2. 设函数 $y = y(x)$ 由参数方程 $\begin{cases} x = \cos t^2 \\ y = t \cos t^2 - \int_0^{t^2} \frac{1}{2\sqrt{u}} \cos u du \end{cases}$ 所确定, 求

$$\frac{dy}{dx}\bigg|_{t=\sqrt{\frac{\pi}{2}}}, \frac{d^2y}{dx^2}\bigg|_{t=\sqrt{\frac{\pi}{2}}}.$$

解 求导得

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\left(t \cos t^2 - \int_0^{t^2} \frac{1}{2\sqrt{u}} \cos u du\right)'}{(\cos t^2)'} = \frac{\cos t^2 + t(-\sin t^2 \cdot 2t) - \frac{1}{2\sqrt{t^2}} \cos t^2 \cdot 2t}{-\sin t^2 \cdot 2t}$$

当 $t > 0$ 时, 有

$$\frac{dy}{dx} = \frac{\cos t^2 + t(-\sin t^2 \cdot 2t) - \frac{1}{2t} \cos t^2 \cdot 2t}{-\sin t^2 \cdot 2t} = \frac{t(-\sin t^2 \cdot 2t)}{-\sin t^2 \cdot 2t} = t$$

于是当 $t > 0$ 时, 有

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dy} \left(\frac{dy}{dt} \right)}{\frac{dx}{dt}} = \frac{(t)'}{(\cos t^2)'} = \frac{1}{-\sin t^2 \cdot 2t} = -\frac{1}{2t \sin t^2}$$

令 $t = \sqrt{\frac{\pi}{2}}$ 得

$$\left. \frac{dy}{dx} \right|_{t=\sqrt{\frac{\pi}{2}}} = t \Big|_{t=\sqrt{\frac{\pi}{2}}} = \sqrt{\frac{\pi}{2}}$$

$$\left. \frac{d^2 y}{dx^2} \right|_{t=\sqrt{\frac{\pi}{2}}} = -\frac{1}{2t \sin t^2} \Big|_{t=\sqrt{\frac{\pi}{2}}} = -\frac{1}{\sqrt{2\pi}}$$

3. 设函数 $y = y(x)$ 由方程 $\begin{cases} x = t^2 + t + 1 \\ \int_1^y e^{u^2} du + t \int_1^t \cos(ut)^2 du = t^3 \end{cases}$ 所确定, 求 $\frac{dy}{dx}$.

解 对第一式关于 t 求导得

$$\frac{dx}{dt} = 2t + 1$$

对于 $\int_1^t \cos(ut)^2 du$, 令 $w = ut$, 则 $u = \frac{w}{t}, du = \frac{1}{t} dw$ (注意 u, w 是积分变量, t 对于

积分是常数), 当 $u = 1$ 时 $w = t$ (积分下限), 当 $u = t$ 时 $w = t^2$ (积分上限), 所以

$$\int_1^t \cos(ut)^2 du = \int_t^{t^2} \cos w^2 \left(\frac{1}{t} dw \right) = \frac{1}{t} \int_t^{t^2} \cos w^2 dw$$

代入第二式得

$$\int_1^y e^{u^2} du + \int_t^{t^2} \cos w^2 dw = t^3 \quad (\text{注意此方程确定隐函数 } y = y(t))$$

两边关于 t 求导

$$\left(\int_1^y e^{u^2} du \right)' + \left(\int_t^{t^2} \cos w^2 dw \right)' = (t^3)'$$

可得

$$e^{y^2} \cdot \frac{dy}{dt} + \cos t^4 \cdot 2t - \cos t^2 = 3t^2 \Rightarrow \frac{dy}{dt} = \frac{3t^2 + \cos t^2 - 2t \cos t^4}{e^{y^2}}$$

因此

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{3t^2 + \cos t^2 - 2t \cos t^4}{e^{y^2}}}{2t+1} = \frac{3t^2 + \cos t^2 - 2t \cos t^4}{(2t+1)e^{y^2}}$$

4. 设 $f(y)$ 是连续函数, 且 $F(x) = \int_a^b f(y)|x-y|dy, a < x < b$, 求 $F''(x)$.

解 (求导去积分号, 必须去掉被积函数中的 x)

$$\begin{aligned} F(x) &= \int_a^b f(y)|x-y|dy = \int_a^x f(y)(x-y)dy + \int_x^b f(y)(y-x)dy \\ &= x \int_a^x f(y)dy - \int_a^x f(y)ydy + \int_x^b f(y)ydy - x \int_x^b f(y)dy \end{aligned}$$

求导得

$$\begin{aligned} F'(x) &= \left[x \int_a^x f(y)dy - \int_a^x f(y)ydy + \int_x^b f(y)ydy - x \int_x^b f(y)dy \right] \\ &= \int_a^x f(y)dy + xf(x) - f(x)x + (-f(x)x) - \int_x^b f(y)dy - x(-f(x)) \\ &= \int_a^x f(y)dy - \int_x^b f(y)dy \\ F''(x) &= \left[\int_a^x f(y)dy - \int_x^b f(y)dy \right]' = f(x) - (-f(x)) = 2f(x) \end{aligned}$$

5. 设函数 $f(x)$ 连续, 求 $\frac{d}{dx} \int_0^x tf(x^2 - t^2)dt$.

解 (求导去积分号, 必须去掉被积函数中的 x)

令 $u = x^2 - t^2$, 则 $du = -2tdt \Rightarrow tdt = -\frac{1}{2}du$, 当 $t = 0$ 时, $u = x^2$ (积分下限),

当 $t = x$ 时, $u = 0$ (积分上限), 于是

$$\int_0^x tf(x^2 - t^2)dt = \int_{x^2}^0 f(u) \left(-\frac{1}{2}du \right) = \frac{1}{2} \int_0^{x^2} f(u)du$$

求导得

$$\frac{d}{dx} \int_0^x tf(x^2 - t^2)dt = \frac{d}{dx} \left[\frac{1}{2} \int_0^{x^2} f(u)du \right] = \frac{1}{2} f(x^2) \cdot 2x = xf(x^2)$$

或

设 $F'(x) = f(x)$, 则

$$\begin{aligned} \int_0^x tf(x^2 - t^2)dt &= -\frac{1}{2} \int_0^x f(x^2 - t^2)d(x^2 - t^2) \\ &= -\frac{1}{2} F(x^2 - t^2) \Big|_{t=0}^{t=x} = -\frac{1}{2} F(0) + \frac{1}{2} F(x^2) \end{aligned}$$

求导得

$$\frac{d}{dx} \int_0^x t f(x^2 - t^2) dt = \frac{d}{dx} \left[-\frac{1}{2} F(0) + \frac{1}{2} F(x^2) \right] = 0 + \frac{1}{2} F'(x^2) \cdot 2x = x f(x^2)$$

6. 设函数 $f(x)$ 连续, 且 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = A$ (A 为实常数), $\varphi(x) = \int_0^1 f(xt) dt$, 求 $\varphi'(x)$,

并讨论 $\varphi'(x)$ 在 $x=0$ 处的连续性.

解 (求导需去掉被积函数中的 x)

当 $x \neq 0$ 时, 令 $u = xt$, 则 $du = x dt$, $dt = \frac{1}{x} du$, 当 $t=0$ 时 $u=0$ (积分下限), 当 $t=1$ 时 $u=x$ (积分上限), 所以

$$\varphi(x) = \int_0^1 f(xt) dt = \int_0^x f(u) \frac{du}{x} = \frac{\int_0^x f(u) du}{x}$$

求导得

$$\varphi'(x) = \left(\frac{\int_0^x f(u) du}{x} \right)' = \frac{xf(x) - \int_0^x f(u) du}{x^2}$$

又由 $f(x)$ 连续及 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = A$ 得 $f(0) = 0$, 则有 $\varphi(0) = \int_0^1 f(0) dt = 0$, 故当

$x=0$ 时, 由导数定义得

$$\varphi'(0) = \lim_{x \rightarrow 0} \frac{\varphi(x) - \varphi(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\frac{\int_0^x f(u) du}{x} - 0}{x} = \lim_{x \rightarrow 0} \frac{\int_0^x f(u) du}{x^2} = \lim_{x \rightarrow 0} \frac{f(x)}{2x} = \frac{A}{2}$$

综上所述可得

$$\varphi'(x) = \begin{cases} \frac{xf(x) - \int_0^x f(u) du}{x^2}, & x \neq 0 \\ \frac{A}{2}, & x = 0 \end{cases}$$

因为

$$\lim_{x \rightarrow 0} \varphi'(x) = \lim_{x \rightarrow 0} \frac{xf(x) - \int_0^x f(u) du}{x^2} = \lim_{x \rightarrow 0} \frac{f(x)}{x} - \lim_{x \rightarrow 0} \frac{\int_0^x f(u) du}{x^2} = A - \frac{A}{2} = \frac{A}{2} = \varphi'(0)$$

所以 $\varphi'(x)$ 在 $x=0$ 处连续.

7. 计算下列极限

$$(1) \quad \lim_{x \rightarrow 0} \frac{\int_0^{x^2} (1 - \sin 2t)^{\frac{1}{t}} dt}{(\mathbf{e}^x - 1) \ln(1 + \arctan x)} ;$$

解

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\int_0^{x^2} (1 - \sin 2t)^{\frac{1}{t}} dt}{(\mathbf{e}^x - 1) \ln(1 + \arctan x)} &= \lim_{x \rightarrow 0} \frac{\int_0^{x^2} (1 - \sin 2t)^{\frac{1}{t}} dt}{x^2} = \lim_{x \rightarrow 0} \frac{\left(\int_0^{x^2} (1 - \sin 2t)^{\frac{1}{t}} dt \right)'}{(x^2)'} \\ &= \lim_{x \rightarrow 0} \frac{(1 - \sin(2x^2))^{\frac{1}{x^2}} \cdot 2x}{2x} = \lim_{x \rightarrow 0} \left[(1 - \sin(2x^2))^{\frac{1}{\sin(2x^2)}} \right]^{\frac{\sin(2x^2)}{x^2}} = \mathbf{e}^{-\lim_{x \rightarrow 0} \frac{\sin(2x^2)}{x^2}} = \mathbf{e}^{-2} \end{aligned}$$

$$(2) \quad \lim_{x \rightarrow 0} \left(\frac{1 + \int_0^x \mathbf{e}^{t^2} dt}{\mathbf{e}^x - 1} - \frac{1}{\sin x} \right) ;$$

解

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1 + \int_0^x \mathbf{e}^{t^2} dt}{\mathbf{e}^x - 1} - \frac{1}{\sin x} \right) &= \lim_{x \rightarrow 0} \frac{(1 + \int_0^x \mathbf{e}^{t^2} dt) \sin x - \mathbf{e}^x + 1}{(\mathbf{e}^x - 1) \sin x} \\ &= \lim_{x \rightarrow 0} \frac{\int_0^x \mathbf{e}^{t^2} dt \cdot \sin x + \sin x - \mathbf{e}^x + 1}{x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \lim_{x \rightarrow 0} \frac{\int_0^x \mathbf{e}^{t^2} dt}{x} + \lim_{x \rightarrow 0} \frac{\sin x - \mathbf{e}^x + 1}{x^2} \\ &= 1 \cdot \lim_{x \rightarrow 0} \frac{\mathbf{e}^{x^2}}{1} + \lim_{x \rightarrow 0} \frac{\cos x - \mathbf{e}^x}{2x} = 1 + \lim_{x \rightarrow 0} \frac{-\sin x - \mathbf{e}^x}{2} = 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

$$(3) \quad \lim_{x \rightarrow 0} \left(1 + \int_0^{\sin x} \sin(t^3) dt \right)^{\frac{1}{x^4 + \ln(1+x^4)}} ;$$

解

$$\begin{aligned} \lim_{x \rightarrow 0} \left(1 + \int_0^{\sin x} \sin(t^3) dt \right)^{\frac{1}{x^4 + \ln(1+x^4)}} &= \lim_{x \rightarrow 0} \left[\left(1 + \int_0^{\sin x} \sin(t^3) dt \right)^{\frac{1}{\int_0^{\sin x} \sin(t^3) dt}} \right]^{\frac{\int_0^{\sin x} \sin(t^3) dt}{x^4 + \ln(1+x^4)}} \\ &= \mathbf{e}^{\lim_{x \rightarrow 0} \frac{\int_0^{\sin x} \sin(t^3) dt}{x^4 + \ln(1+x^4)}} = \mathbf{e}^{\lim_{x \rightarrow 0} \frac{\int_0^{\sin x} \sin(t^3) dt}{2x^4}} = \mathbf{e}^{\lim_{x \rightarrow 0} \frac{\sin(\sin^3 x) \cos x}{8x^3}} = \mathbf{e}^{\lim_{x \rightarrow 0} \frac{x^3}{8x^3}} = \mathbf{e}^{\frac{1}{8}} \end{aligned}$$

$$(4) \quad \lim_{x \rightarrow 0} \frac{\int_0^x \left(\int_0^{u^2} \arctan(1+t) dt \right) du}{x(1 - \cos x)} ;$$

解

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\int_0^x \left(\int_0^{u^2} \arctan(1+t) dt \right) du}{x(1-\cos x)} &= \lim_{x \rightarrow 0} \frac{\int_0^x \left(\int_0^{u^2} \arctan(1+t) dt \right) du}{\frac{x^3}{2}} = \lim_{x \rightarrow 0} \frac{\left[\int_0^x \left(\int_0^{u^2} \arctan(1+t) dt \right) du \right]'}{\left(\frac{x^3}{2} \right)'} \\ &= \lim_{x \rightarrow 0} \frac{\int_0^{x^2} \arctan(1+t) dt}{\frac{3x^2}{2}} = \lim_{x \rightarrow 0} \frac{\left[\int_0^{x^2} \arctan(1+t) dt \right]'}{\left(\frac{3x^2}{2} \right)'} = \lim_{x \rightarrow 0} \frac{\arctan(1+x^2) \cdot 2x}{3x} \\ &= \frac{2}{3} \lim_{x \rightarrow 0} \arctan(1+x^2) = \frac{2}{3} \arctan 1 = \frac{2}{3} \cdot \frac{\pi}{4} = \frac{\pi}{6} \end{aligned}$$

$$(5) \lim_{x \rightarrow 0} \frac{\int_1^{e^x} \sin(e^x - t)^2 dt}{x^2 \ln(1+x)};$$

解 (求导去积分号, 必须去掉被积函数中的 x)

令 $u = e^x - t$, 则 $t = e^x - u, dt = -du$ (注意, t, u 是积分变量, 在积分中 x 看成常数),

当 $t = 1$ 时 $u = e^x - 1$ (积分下限), 当 $t = e^x$ 时 $u = 0$ (积分上限), 所以

$$\int_1^{e^x} \sin(e^x - t)^2 dt = \int_{e^x-1}^0 \sin u^2 (-du) = \int_0^{e^x-1} \sin u^2 du$$

于是

$$\lim_{x \rightarrow 0} \frac{\int_1^{e^x} \sin(e^x - t)^2 dt}{x^2 \ln(1+x)} = \lim_{x \rightarrow 0} \frac{\int_0^{e^x-1} \sin u^2 du}{x^3} = \lim_{x \rightarrow 0} \frac{\sin(e^x - 1)^2 \cdot e^x}{3x^2} = \lim_{x \rightarrow 0} \frac{x^2}{3x^2} = \frac{1}{3}$$

$$(6) \lim_{x \rightarrow 0^+} \frac{\int_0^x (\sqrt{x-t})(e^t) dt}{\sqrt{x^3}}.$$

解 (求导去积分号, 必须去掉被积函数中的 x)

令 $u = x - t$, 则 $t = x - u, dt = -du$ (注意, t, u 是积分变量, 在积分中 x 看成常数),

当 $t = 0$ 时 $u = x$ (积分下限), 当 $t = x$ 时 $u = 0$ (积分上限), 所以

$$\int_0^x (\sqrt{x-t})(e^t) dt = \int_x^0 \sqrt{u} e^{x-u} (-du) = \int_0^x \sqrt{u} e^{x-u} du = \int_0^x \sqrt{u} e^x e^{-u} du = e^x \int_0^x \sqrt{u} e^{-u} du$$

于是

$$\begin{aligned}\lim_{x \rightarrow 0^+} \frac{\int_0^x (\sqrt{x-t})(e^t) dt}{\sqrt{x^3}} &= \lim_{x \rightarrow 0^+} \frac{e^x \int_0^x \sqrt{u} e^{-u} du}{\sqrt{x^3}} = \lim_{x \rightarrow 0^+} e^x \lim_{x \rightarrow 0^+} \frac{\int_0^x \sqrt{u} e^{-u} du}{\sqrt{x^3}} \\ &= e^0 \lim_{x \rightarrow 0^+} \frac{\left(\int_0^x \sqrt{u} e^{-u} du \right)'}{\left(\sqrt{x^3} \right)'} = \lim_{x \rightarrow 0^+} \frac{\sqrt{x} e^{-x}}{\frac{3}{2} \sqrt{x}} = \frac{2}{3} \lim_{x \rightarrow 0^+} e^{-x} = \frac{2}{3} e^0 = \frac{2}{3}\end{aligned}$$

8. 设 $f(x)$ 可导, 且 $f(0) = 0, f'(0) = 0$, 求 $\lim_{x \rightarrow 0} \frac{\int_0^x t f(x^2 - t^2) dt}{(1 - \cos x)(\sqrt{1 + x^2} - 1)}$.

解 (求导去积分号, 必须去掉被积函数中的 x)

$$\int_0^x t f(x^2 - t^2) dt = -\frac{1}{2} \int_0^x f(x^2 - t^2) (x^2 - t^2)' dt = -\frac{1}{2} \int_0^x f(x^2 - t^2) d(x^2 - t^2) \quad \text{令}$$

$u = x^2 - t^2$, 则 $d(x^2 - t^2) = du$, 当 $t = 0$ 时 $u = x^2$ (积分下限), 当 $t = x$ 时 $u = 0$ (积分上限), 所以

$$\int_0^x t f(x^2 - t^2) dt = -\frac{1}{2} \int_{x^2}^0 f(u) du = \frac{1}{2} \int_0^{x^2} f(u) du$$

于是

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\int_0^x t f(x^2 - t^2) dt}{(1 - \cos x)(\sqrt{1 + x^2} - 1)} &= \lim_{x \rightarrow 0} \frac{\frac{1}{2} \int_0^{x^2} f(u) du}{\frac{x^2}{2} \cdot \frac{x^2}{2}} = 2 \lim_{x \rightarrow 0} \frac{\int_0^{x^2} f(u) du}{x^4} \\ &= 2 \lim_{x \rightarrow 0} \frac{\left(\int_0^{x^2} f(u) du \right)'}{(x^4)'} = 2 \lim_{x \rightarrow 0} \frac{f(x^2) \cdot 2x}{4x^3} = \lim_{x \rightarrow 0} \frac{f(x^2)}{x^2} = \lim_{x \rightarrow 0} \frac{f(x^2) - f(0)}{x^2 - 0} = f'(0) = 0\end{aligned}$$

9. 设 $f(x)$ 是区间 $(-\infty, +\infty)$ 内的连续函数, 且 $f(0) \neq 0$, 计算极限

$$\lim_{x \rightarrow 0} \frac{\int_0^x t f(x-t) dt}{\int_0^x x f(x-t) dt}.$$

解 (求导去积分号, 必须去掉被积函数中的 x)

令 $u = x - t$, 则 $t = x - u, dt = -du$, 当 $t = 0$ 时 $u = x$ (积分下限), 当 $t = x$ 时 $u = 0$

(积分上限), 所以

$$\int_0^x t f(x-t) dt = \int_x^0 (x-u) f(u) (-du) = \int_0^x (x-u) f(u) du = x \int_0^x f(u) du - \int_0^x u f(u) du$$

$$\int_0^x xf(x-t)dt = \int_x^0 xf(u)(-du) = \int_0^x xf(u)du = x \int_0^x f(u)du$$

于是

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\int_0^x tf(x-t)dt}{\int_0^x xf(x-t)dt} &= \lim_{x \rightarrow 0} \frac{x \int_0^x f(u)du - \int_0^x uf(u)du}{x \int_0^x f(u)du} \\ &= \lim_{x \rightarrow 0} \frac{\left(x \int_0^x f(u)du - \int_0^x uf(u)du \right)'}{\left(x \int_0^x f(u)du \right)'} = \lim_{x \rightarrow 0} \frac{\int_0^x f(u)du + xf(x) - xf(x)}{\int_0^x f(u)du + xf(x)} \\ &= \lim_{x \rightarrow 0} \frac{\int_0^x f(u)du}{\int_0^x f(u)du + xf(x)} \quad (\text{注意此时不能用求导去积分号了, 因为 } f(x) \text{ 没有导数}) \\ &= \lim_{x \rightarrow 0} \frac{f(\xi)x}{f(\xi)x + xf(x)} \quad (\text{用定积分中值公式, } \xi \text{ 介于 } 0 \text{ 与 } x \text{ 之间, 当 } x \rightarrow 0 \text{ 时 } \xi \rightarrow 0) \\ &= \lim_{x \rightarrow 0} \frac{f(\xi)}{f(\xi) + f(x)} = \frac{f(0)}{f(0) + f(0)} = \frac{1}{2} \end{aligned}$$

10. 设两数列 $\{a_n\}$ 和 $\{b_n\}$ 满足 $a_n > 0, b_n > 0$ 及 $\int_{\sin a_n}^{a_n} e^{x^2} dx = b_n \ln(1+b_n)$, 且

$$\lim_{n \rightarrow \infty} a_n = 0, \quad (1) \quad \text{证明 } \lim_{n \rightarrow \infty} b_n = 0; \quad (2) \quad \text{求 } \lim_{n \rightarrow \infty} \frac{a_n^3}{b_n^2}.$$

证 (1) 因为 $\lim_{n \rightarrow \infty} a_n = 0$, 所以

$$\lim_{n \rightarrow \infty} b_n \ln(1+b_n) = \lim_{n \rightarrow \infty} \int_{\sin a_n}^{a_n} e^{x^2} dx = \int_0^0 e^{x^2} dx = 0$$

从而 $\lim_{n \rightarrow \infty} b_n = 0$.

(2)

$$\lim_{n \rightarrow \infty} \frac{a_n^3}{b_n^2} = \lim_{n \rightarrow \infty} \frac{a_n^3}{b_n \ln(1+b_n)} \cdot \frac{b_n \ln(1+b_n)}{b_n^2} = \lim_{n \rightarrow \infty} \frac{\ln(1+b_n)}{b_n} \lim_{n \rightarrow \infty} \frac{a_n^3}{\int_{\sin a_n}^{a_n} e^{x^2} dx} = \lim_{n \rightarrow \infty} \frac{a_n^3}{\int_{\sin a_n}^{a_n} e^{x^2} dx}$$

因为

$$\lim_{x \rightarrow 0} \frac{x^3}{\int_{\sin x}^x e^{x^2} dx} = \lim_{x \rightarrow 0} \frac{x^3}{e^{\xi^2} (x - \sin x)} = \lim_{x \rightarrow 0} \frac{1}{e^{\xi^2}} \lim_{x \rightarrow 0} \frac{x^3}{x - \sin x} = 1 \cdot \lim_{x \rightarrow 0} \frac{3x^2}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{6x}{\sin x} = 6$$

所以

$$\lim_{n \rightarrow \infty} \frac{a_n^3}{b_n^2} = \lim_{n \rightarrow \infty} \frac{a_n^3}{\int_{\sin a_n}^{a_n} e^{x^2} dx} = 6$$